Does Adaptive Checkpointing Help Spark Structured Streaming?

A Fault-Injection Study of Replay, Recovery and Latency

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https://github.com/rmedipallycic/spark-streaming-fault-tolerance

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Abstract

Checkpoint-interval tuning is usually framed as a trade-off between overhead in normal operation and work lost on failure, and adaptive controllers aim to shorten the interval when failures look likely. We tested whether that framing holds for Spark Structured Streaming, where the micro-batch trigger interval also sets how often the offset and commit logs are written. We built a fault-injection harness that SIGKILLs a Spark job reading sequence-numbered records from Kafka, restarts it from its checkpoint, and counts every duplicated or missing record, and implemented an adaptive controller (ACS) that changes the trigger at runtime by restarting the query between batches. Over 210 measured trials on one machine, a crash caused a batch to be reprocessed in only 9 of 120 comparison trials, because records that arrive between batches are read for the first time after a restart, not replayed; recovery took about 10 s whatever the interval and was dominated by restart. The interval therefore traded end-to-end latency against checkpoint overhead, not against lost work. Interval switches were cheap (median 0.54 s) and safe (no duplicates in 320 switches), but on paired, identical workloads and failures a fixed interval chosen at ACS's own mean matched or beat it: against a fixed 5.8 s interval, the original ACS rule had the lower 95th-percentile latency in 0/10 pairs and fewer commits in only 2/10. The original rule also cannot react to any single signal. We report the negative result, its mechanism, and two measurement pitfalls we hit along the way: an earlier simulation that predicted the wrong failure behaviour, and laptop standby that silently paused six trials, which we detected from the producer's clock.

1 Introduction

Fault-tolerant stream processors periodically record how far they have read and what state they hold, so that after a failure they can resume instead of starting over. How often to do this is a classic question [12, 5]: checkpoint too often and normal operation pays for it; too rarely and a failure throws away more work. Adaptive policies that tighten the interval when the system looks unstable follow naturally from that framing.

We set out to build and evaluate such a policy for Spark Structured Streaming [1]. An earlier simulation of the policy, Adaptive Checkpoint Selection (ACS), suggested large reductions in duplicated records. When we audited that simulation, we found that its advantage followed from how the model was written, and we replaced it with live measurements on real Spark. This paper reports what those measurements showed.

The answer was negative, for a reason that is specific to micro-batch execution and easy to miss. In Structured Streaming the processing-time trigger sets both how often a batch runs and how often progress is written to the checkpoint. Records that arrive between two triggers are not lost or replayed when the driver dies: they are simply read for the first time by the restarted query. Only a batch that is *running* at the moment of the crash is redone. Longer intervals therefore do not mean more lost work in the usual sense; they mean higher latency, because records wait longer for the next batch. The interval is a latency-versus-overhead knob, and a well-chosen fixed value served that trade-off as well as adaptive control did.

Contributions.

1. A fault-injection harness for Spark Structured Streaming with Kafka that counts every duplicated and missing record by sequence number, measures replay from the checkpoint logs, end-to-end latency per record, and recovery time, and records the code commit and environment of every trial.

2. A measured account of delivery under driver failure and checkpoint corruption for the exactly-once file sink and an at-least-once foreachBatch sink (84 trials).

3. A live, restart-based implementation of adaptive trigger control, with its switching cost, and an analysis showing that the original ACS rule cannot react to any single signal.

4. A paired comparison of ACS against fixed intervals, including fixed intervals set to ACS's own mean, in two runs (60 and 66 trials), showing no advantage for adaptation and explaining why.

5. Two methodology lessons: simulated failure behaviour that differed from the measured behaviour in kind, and a clock-based test that detected trials paused by laptop standby.

2 Background

Micro-batch execution. Structured Streaming runs a streaming query as a sequence of small batch jobs [1, 13]. With a processing-time trigger of T seconds, the driver starts a batch every T seconds if new data is available. Before a batch runs, the driver writes the range of source offsets it will read to the checkpoint's `offsets/` log; after the sink has finished, it writes an entry to `commits/`. On restart, Spark finds the latest offsets entry; if it has no matching commit, the batch is run again with the same offsets.

End-to-end guarantees. Exactly-once output needs a replayable source (Kafka offsets [8]), deterministic re-execution of a batch from its recorded offsets, and an idempotent or transactional sink [10]. Spark's file sink provides the last part through its own `_spark_metadata` log, which lists the files that belong to each committed batch. A `foreachBatch` sink hands each batch to user code; if that code is not idempotent, a batch that is run again is written again (at-least-once).

What the interval controls. Because offsets and commits are written once per batch, the trigger interval is also the checkpoint interval. Section 7 shows why, unlike in the classic model, a longer interval does not proportionally increase the work redone after a crash.

3 Harness and setup

One trial. A trial creates a fresh Kafka topic and produces 36,000 records with sequence numbers through an idempotent producer, so that any duplicate is Spark's. It starts the Spark job from the earliest offset, waits for the first committed output, injects the failure 15 s plus a per-trial phase later, restarts the job from the same checkpoint, waits until output stops changing, and stops the query gracefully. It then counts duplicated and missing sequence numbers in the committed output and, separately, in every output file. Each trial writes a JSON record with its configuration, git commit (and whether the working tree was dirty; 0 trials were run from a dirty tree), package versions, event timestamps and counts, plus the Spark log, the controller log and a checkpoint snapshot taken before the restart.

Failures. *Driver kill*: SIGKILL of the Spark process group (in local mode, driver and executors are one JVM), then restart. *Offset-log corruption*: the same kill, then random bytes written over the middle of the latest file in `offsets/`, then restart.

Metrics. *Recovery*: time from the kill to the first new entry in `commits/` after the restart. *Replay*: records in batches that had an offsets entry but no commit at the kill, read from the checkpoint before the restart. *End-to-end latency*: per record, the modification time of the first output file containing it minus the producer's send time; we report each trial's median (p50) and 95th percentile (p95). *Batch commits*: the number of committed micro-batches in the trial, our measure of checkpoint overhead (every trial processes the same 36,000 records).

Environment. One laptop (Intel Core i5-1335U) running WSL2 Ubuntu with 8 GB for WSL; Spark Structured Streaming 3.5.3 through PySpark in `local[2]` mode; Java 17; one Kafka broker (Confluent 7.5.0) in Docker Compose, with a 4-partition topic per trial. Everything ran on the same machine, so network and cluster scheduling play no part; Section 9 discusses what this does and does not allow us to conclude.

4 Study 1: what Spark delivers under failure

The first study used two fixed triggers, A (1 s) and B (30 s), both sinks, both failures, and 10 trials per cell plus one failure-free baseline each, at a constant 300 records/s (84 trials, code commits 4a2dfe3, 5607d08; an earlier attempt at B, invalidated by a full `/tmp`, is kept and documented in the repository). Kill times were spread across the trigger interval by a golden-ratio sequence.

Table 1 shows three results. The file sink delivered every record exactly once in all 20 driver kills; even the uncommitted files contained no duplicates. The at-least-once foreachBatch sink duplicated records in one trial: 300 records, exactly one 1-second batch at 300 records/s, written again after the restart. A duplicate requires the kill to land between a batch's write and its commit, a short window that none of the ten 30 s kills hit. Offset-log corruption stopped the pipeline in 40 of 40 trials: Spark refused to restart (`MalformedInputException`, exit code 3) instead of continuing with wrong offsets, so no records were du-

Table 1: Study 1: driver kill and offset-log corruption (10 trials per cell; baselines: 1). Missing records are counted only for trials whose restart succeeded. Recovery: median (min–max), seconds.

Failure	Sink	Trigger	Trials	Outcomes	Trials w/ dup.	Max dup. rows	Missing	Recovery s
driver kill	file	A (1 s)	10	ok 10	0	0	0/10	11.0 (9.5–15.5)
driver kill	file	B (30 s)	10	ok 10	0	0	0/10	14.8 (12.0–17.0)
driver kill	foreachBatch	A (1 s)	10	ok 10	1	300	0/10	10.0 (9.5–28.6)
driver kill	foreachBatch	B (30 s)	10	ok 10	0	0	0/10	13.3 (11.0–20.5)
offset-log corruption	file	A (1 s)	10	restart failed 10	0	0	0/0	–
offset-log corruption	file	B (30 s)	10	restart failed 10	0	0	0/0	–
offset-log corruption	foreachBatch	A (1 s)	10	restart failed 10	0	0	0/0	–
offset-log corruption	foreachBatch	B (30 s)	10	restart failed 10	0	0	0/0	–
none (baseline)	file	A (1 s)	1	ok 1	0	0	0/1	–
none (baseline)	file	B (30 s)	1	ok 1	0	0	0/1	–
none (baseline)	foreachBatch	A (1 s)	1	ok 1	0	0	0/1	–
none (baseline)	foreachBatch	B (30 s)	1	ok 1	0	0	0/1	–

Table 2: Driver failure: earlier simulation versus Study 1. Duplicate rate: duplicated rows as a share of records, mean over trials; fBatch: foreachBatch sink. Recovery: simulated mean versus measured median, both sinks.

	Simulated		Measured		
Trigger	dup. %	rec. s	file	fBatch	rec. s
A (1 s)	0.19	5.1	0.00	0.083	10.5
B (30 s)	4.43	25.3	0.00	0.000	14.3

plicated or silently lost; processing simply stopped until an operator intervened.

Recovery after a driver kill took 9.5–28.6 s. From the timestamps in each Spark log, a median 67% (at least 57%, over 40 kills) of that time passed before the restarted query was running, i.e. in JVM and Spark start-up.

Simulation versus measurement. Table 2 compares these measurements with the earlier simulation of the same strategies. The simulation assigned every failure a constant duplicate rate, including 0.18% (A) and 2.96% (B) for checkpoint corruption, where the measured pipeline instead stopped. It had no model of a sink, so it could not represent the file sink, which admits no duplicates by design, and it predicted recovery growing about fivefold from A to B. Measured, B recovered a few seconds slower than A in this study, but A and B ran on different days; when every strategy was interleaved (Section 6), recovery did not depend on the interval. The differences are in kind, not degree, and they are why this paper reports only measured results.

5 Adaptive checkpoint selection, live

The rule. ACS evaluates every 5 s a risk score $r = 0.4\,u + 0.3\,e + 0.3\,f$ from three signals in $[0, 1]$: throughput instability u (coefficient of variation of input rate over the last ten windows, saturating at 0.3; 0.2 during warm-up), error pressure e, and failure recency $f = \max(0, 1 - t_{\text{since failure}}/120\,\text{s})$. If $r > 0.45$ the interval shrinks by 30%; otherwise the window counts as stable, and after four stable windows the interval grows by 20%, within 2–45 s, starting at 10 s. In the simulation, e counted exceptions per 1,000 records. A healthy live pipeline has none, so the live controller measures e as the fraction of batches in the window whose run time exceeded the trigger interval, a measurable sign of falling behind.

A structural flaw (v1). Implementing the rule exposed a property the simulation had hidden: no single signal can cross the threshold. Instability alone contributes at most 0.4, and a recent failure or full error pressure at most 0.38 (with warm-up instability). And because every window with $r \leq 0.45$ counts as stable, the interval *relaxes* during a burst unless a failure happened within the previous 100 s. We therefore evaluate two variants, both fixed before any measured run: **v1**, the rule exactly as specified, and **v2**, which tightens when $r > 0.30$ (any one saturated signal), counts a window as stable only when $r < 0.15$, and otherwise holds the interval.

Changing the interval of a running query. A running Structured Streaming query cannot change its trigger. Our controller runs inside the Spark job and, to apply a new interval, waits until the query is between batches, stops it, and starts it again on the same checkpoint in the same Spark session, so no JVM restart is involved. Waiting for the gap between batches matters: stopping a query interrupts a running batch, and for foreachBatch a batch that had already written its output would be written again. The controller persists its interval and a clean-stop flag next to the checkpoint, so that after a crash the restarted job resumes the interval and counts the crash for the recency signal.

Switching cost. Over 320 switches in 40 ACS trials, the time from stopping the old query to the first batch of the new one had a median of 0.54 s (0.52–2.07 s; 33 of 318 measured switches took more than 1 s), against about 10 s for crash recovery. No ACS trial produced a duplicate (0 trials with duplicates). Restart-based adaptation is therefore cheap and safe; whether it is *useful* is the question of the next two sections.

6 Study 2: ACS against fixed intervals

Design. Six strategies: fixed triggers of 1 s (A) and 30 s (B), ACS v1 and v2, and two further fixed triggers set to ACS's mean intervals from the first run, 3.3 s and 5.8 s (C). The C strategies are the fair comparison: they answer whether *adapting* helps, rather than merely choosing a value between 1 s and 30 s. The workload was bursty: every 15 s the producer's rate was drawn from 0.5×, 1× and 3× 300 records/s, from a seed equal to the trial number. All trials used the foreachBatch sink and a driver kill at 15 s plus a phase spread over 30 s, the same for every strategy. Trial t therefore gave every strategy the same input and the same kill time, so strategies can be compared pair by pair.

Two runs. The first run (60 trials) ran A, B and both ACS variants interleaved overnight, and the two C strategies the next morning, separately from the others. The 5.8 s C trials recovered more slowly (median 14.3 s against about 10 s), and their Spark start-up was slower too, which no trigger setting can cause, so the first run's comparison with C is biased against C. The second run (66 trials) interleaved all six strategies trial by trial and is our main evidence. Six of its trials were paused by laptop standby (Section 8); they are kept in the data, excluded from the tables, and were rerun with the same trial numbers, giving 60 clean trials, 10 per strategy. Code commits: 369028b, 3d1f93f, 7d66520.

Results. Table 3 and Figure 1 show the second run. The fixed intervals trace a clean trade-off: from A to B, median latency rises from 0.84 s to 13.98 s while batch commits fall from 82 to 6. The question is where ACS lies relative to that curve.

ACS v1 (mean interval 5.8 s) had a median latency of 3.68 s with 24 commits; the fixed curve at the same number of commits gives 2.73 s. Against the fixed 5.8 s interval, it had a higher p95 latency (10.88 versus 8.11 s) and more commits (24 versus 18). Pair by pair (Table 4), v1 had the lower median latency in 5/10 pairs, the lower p95 in 0/10, and fewer commits in 2/10.

ACS v2 (mean interval 3.5 s) had a median latency of 1.99 s with 35 commits, essentially on the fixed curve

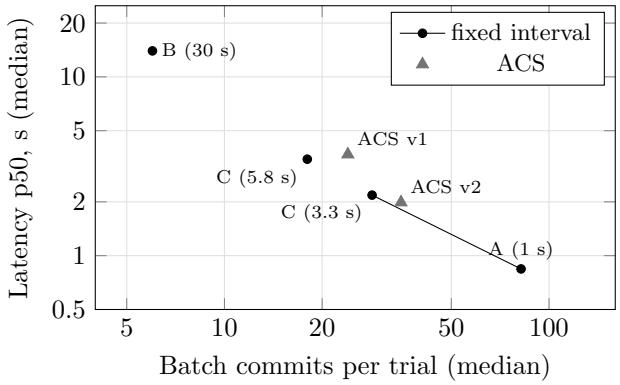


Figure 1: Latency against checkpoint overhead, second run. The line joins the fixed intervals (B, C 5.8 s, C 3.3 s, A from left to right). ACS v2 lies on the line; ACS v1 lies above it.

(2.02 s at that number of commits). Against the fixed 3.3 s interval it had a slightly lower median latency (1.99 versus 2.18 s; lower in 7/10 pairs), a higher p95 (8.89 versus 8.61 s; lower in only 1/10) and more commits (35 versus 28; fewer in only 1/10).

The first run (Table 5) shows the same ordering for A, B and the two ACS variants. Its C strategies ran separately, and the 5.8 s one had higher latency than in the second run, which made ACS v1 look better against it (9/10 pairs with lower median latency); even so, every ACS variant used more commits than the fixed interval at its own mean in both runs.

7 Why adaptation has nothing to gain

Replay is rare. In the classic model, a failure loses the work done since the last checkpoint, so a longer interval means more redone work. In micro-batch Spark, records that arrived since the last batch were never processed; after a restart they are read for the first time. Only a batch that was running when the driver died is run again. Across both runs, that happened in 9 of 120 clean trials (Table 6).

When it happens, it scales with the batch. The replayed amount is the size of the interrupted batch, so it grows with the interval and the current rate: the largest, 15,750 records, came from a single 30 s batch. Roughly, the chance of killing a batch in flight is its run time divided by the interval, while its size grows with the interval, so the expected replay depends much less on the interval than the classic model suggests. Our ten trials per strategy are too few to estimate that expectation precisely; we report the events individually.

Table 3: Study 2, second (interleaved) run: per-trial medians, then median (min–max) over 10 trials. Latency: producer send to output. Batch commits: committed micro-batches per trial (36,000 records each). Mean interval: time-weighted, ACS only.

Strategy	Trials	Latency p50 s	Latency p95 s	Batch commits	Mean interval s	Switches	Recovery s
A (1 s)	10	0.84 (0.71–1.38)	8.75 (3.79–14.39)	82 (43–101)	–	–	10.5 (9.6–16.6)
C (3.3 s)	10	2.18 (1.91–2.91)	8.61 (4.29–10.83)	28 (17–40)	–	–	10.5 (8.5–14.1)
ACS v2	10	1.99 (1.43–4.36)	8.89 (6.68–12.15)	35 (14–56)	3.5 (2.9–4.3)	6 (5–6)	10.3 (9.5–14.7)
C (5.8 s)	10	3.46 (3.25–4.27)	8.11 (5.97–11.61)	18 (10–24)	–	–	10.0 (9.5–15.0)
ACS v1	10	3.68 (1.81–6.40)	10.88 (9.29–13.30)	24 (8–34)	5.8 (3.1–6.4)	10 (8–13)	10.3 (9.5–14.1)
B (30 s)	10	13.98 (9.16–24.36)	30.44 (25.76–46.45)	6 (3–6)	–	–	10.5 (10.0–25.6)

Table 4: Paired comparison: in how many of the 10 pairs (same input, same kill time) ACS had the lower value.

Run	Comparison	p50	p95	Commits
1	ACS v2 vs C (3.3 s)	6/10	3/10	0/10
1	ACS v1 vs C (5.8 s)	9/10	7/10	2/10
2	ACS v2 vs C (3.3 s)	7/10	1/10	1/10
2	ACS v1 vs C (5.8 s)	5/10	0/10	2/10

Table 5: Both runs, medians over clean trials. Run 1's C strategies ran separately from the others (see text).

	Run 1			Run 2		
Strategy	p50	p95	comm.	p50	p95	comm.
A (1 s)	0.76	6.10	84	0.84	8.75	82
C (3.3 s)	2.12	7.01	30	2.18	8.61	28
ACS v2	1.73	8.36	37	1.99	8.89	35
C (5.8 s)	3.96	11.92	17	3.46	8.11	18
ACS v1	3.48	10.27	26	3.68	10.88	24
B (30 s)	12.65	29.23	6	13.98	30.44	6

Replay did not lengthen recovery, and rarely duplicated. Recovery in the replay trials was 9.0–14.0 s, within the range of all other trials; the 15,750-record replay recovered in 10.5 s. Only one of the nine replays produced duplicates (1,739 rows): the interrupted batch had already written its output, the at-least-once window from Study 1. In the others the batch died before its write and was simply redone.

Table 6: Every trial in which the kill interrupted a running batch (replay > 0), both runs. Dup.: duplicated rows in the output.

Run	Strategy	Trial	Replayed	Recovery s	Dup.
1	ACS v2	2	89	9.5	0
1	A (1 s)	3	150	9.0	0
1	A (1 s)	8	150	9.0	0
1	ACS v2	4	236	9.5	0
1	ACS v2	9	490	10.0	0
1	C (5.8 s)	7	1,739	14.0	1,739
2	A (1 s)	7	301	10.5	0
2	ACS v2	7	981	10.0	0
2	B (30 s)	1	15,750	10.5	0

Table 7: Recovery split, second run: Spark start-up after the restart, and the first batch after start-up. Median (min–max), seconds.

Strategy	n	Start-up	First batch
A (1 s)	10	7.2 (6.5–10.7)	3.5 (2.7–5.8)
C (3.3 s)	10	7.2 (6.1–8.9)	3.3 (2.4–6.3)
ACS v2	10	7.0 (6.2–11.0)	3.5 (2.7–4.3)
C (5.8 s)	10	6.7 (6.3–9.8)	3.4 (2.8–5.2)
ACS v1	10	7.2 (6.3–9.9)	3.4 (2.9–5.0)
B (30 s)	10	7.0 (6.5–18.0)	3.6 (3.0–7.5)

Recovery does not depend on the interval. Median recovery was 10.0–10.5 s for every strategy in the second run (Table 3). Splitting each recovery at the moment the restarted query was running (Table 7), start-up took about 7 s and the first batch after it about 3.5 s, for every strategy. Over both runs, 90 of 120 recoveries took under 12 s and 29 took 12–17 s; the slower ones had both slower start-up (median 9.1 s against 6.7 s) and a slower first batch (5.0 against 3.3 s), which points to the machine's state at the time, not to the strategy. One recovery took 25.6 s, with an 18 s start-up.

What is left is latency against overhead. With replay rare and recovery independent of the interval, the interval governs how long records wait for a batch (latency) and how many batches commit (overhead). That trade-off depends on the workload's rate and the latency target, both of which an operator knows in advance. A fixed interval chosen for the target serves it directly. ACS adds switching (cheap, but not free), and its decisions follow failure recency and throughput variance, which are signals for the risk of losing work, the very cost shown above to be small here.

Table 8: Stalled trials (producer more than 10 s behind schedule), excluded from the tables and rerun with the same trial numbers.

Run id (start, UTC)	Strategy	Trial	Overrun s
20261016T223348Z	C (3.3 s)	0	2431
20261017T000019Z	B (30 s)	2	575
20261017T003005Z	C (3.3 s)	3	376
20261017T005701Z	A (1 s)	4	264
20261017T010347Z	B (30 s)	4	684
20261017T012115Z	C (5.8 s)	4	1691

8 Detecting paused trials

Six trials of the second run took minutes longer than any other, with p95 latencies up to 536 s. Their outcomes were correct; the timings were not Spark's. The producer is paced only by the wall clock, so the time it needs to send all records follows from the workload schedule. We define a trial as *stalled* if its producer finished more than 10 s later than its schedule allows. Over 126 trials, 120 finished within 1 s of schedule, none between 1 and 10 s, and 6 were 4 to 40 minutes late (Table 8). The Windows power log showed that the laptop had entered Modern Standby after an idle timeout during each of those periods, although it was plugged in; an excerpt is in the repository. The test needs no access to the operating system and catches a pause wherever it falls in a trial; checking only the time to first output would have missed five of the six.

9 Threats to validity

- **One machine, local mode.** Driver and executors share one JVM, so executor-only failures, network partitions and multi-node scheduling are not covered. On a cluster, batches take longer to schedule and the in-flight window is larger, which could make replay more frequent; the mechanism in Section 7 still applies, but its magnitude may differ.
- **One workload, low rate.** 300 records/s with 0.5–3× bursts and a trivial transformation. Heavier batches spend more time in flight, which raises the chance that a kill interrupts one.
- **Ten trials per strategy and run.** Enough to show the ordering and its consistency across pairs and runs, not to estimate small differences or rare-event rates. Replay events are reported individually for that reason.
- **Machine state.** Recovery showed two modes tied to the machine's state, and the first run's C strategies ran under different conditions. The second run interleaves every strategy trial by trial; its six reruns ran separately and recovered in the slower mode.
- **One adaptive rule.** We tested ACS as specified and one revision. A controller that targets latency directly, or tunes batch size as in [6], is a different question; our results bound what an interval controller driven by failure risk can gain here, not every possible controller.
- **One kind of corruption.** Overwriting part of the latest offsets file; other corruptions (commit log, sink metadata, truncation) may behave differently.

10 Related work

Discretized streams [13] introduced micro-batch stream processing with lineage-based recovery, and Structured Streaming [1] its declarative successor with exactly-once sinks. Das et al. [6] adapt the batch interval of Spark Streaming to keep the system stable as load changes, a latency and stability goal rather than a fault-tolerance one, consistent with our finding that the interval is a latency knob. Drizzle [11] decouples the processing interval from coordination to cut micro-batch overhead. Flink takes consistent snapshots with barriers derived from Chandy–Lamport [4, 3, 2], where the checkpoint interval and the processing granularity are separate and the classic trade-off applies more directly. Optimal checkpoint intervals under failure were derived by Young [12] and Daly [5], and rollback recovery is surveyed by Elnozahy et al. [7]. The same measurement approach, a fault-injection harness with committed raw logs, was applied to LLM serving in [9].

11 Conclusion

In Spark Structured Streaming on one machine, a driver crash rarely forced a batch to be redone (9 of 120 trials), because records that arrive between batches are read after the restart, not replayed. Recovery took about 10 s regardless of the trigger interval, mostly in restart. The interval therefore traded latency against checkpoint overhead. Changing it at runtime was cheap and safe, but on paired workloads and failures, a fixed interval at ACS's own mean matched or beat it, and the original ACS rule could not react to any single signal. For micro-batch Spark, our evidence supports choosing a fixed trigger for the latency target over adapting it to failure risk. Next steps: a multi-node cluster with executor failures and heavier batches, where in-flight batches are larger; other corruption modes; and controllers that target latency directly.

Artifact. Code, every trial record and log, and the script that generates every table, figure and in-text number of this paper are at `https://github.com/rmedipallycic/spark-streaming-fault-tolerance` (MIT licence). Excluded trials are kept with their reasons.

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